

PROPOSED TRAINING PROGRAMME

Industry-Oriented Training on Structural Engineering and Geospatial Applications Using AutoCAD, SAP2000 and QGIS

1. Programme Summary

Particulars	Details
Programme Title	Industry-Oriented Training on Structural Engineering and Geospatial Applications using AutoCAD, SAP2000 and QGIS
Target Participants	II and III Year B.Tech/B.E. Civil Engineering Students
Batch Size	60 Students
Total Duration	60 Hours
Direct Training Hours	30 Hours
Reading Material and Assignment Hours	30 Hours
Training Period	30 Days
Daily Engagement	Around 2 hours per day: 1-hour direct session + 1 hour reading/ practice/ assignment
Mode of Delivery	Online for AutoCAD and SAP2000; offline practical sessions only for QGIS
Software Platforms	AutoCAD, SAP2000 and QGIS
Final Submission	Group-based project report only; presentation not required

2. Resource Persons

The programme will be delivered by resource persons with software-specific responsibilities. The sessions will be coordinated to maintain continuity across drafting, structural modelling and geospatial applications.

Module / Software	Resource Person	Role in the Training
AutoCAD: Structural Drafting and Detailing	Ms. Neha Anil Choudhari	Resource person for AutoCAD sessions covering drawing setup, layers, structural layouts, detailing conventions and drawing documentation.
SAP2000: Structural Analysis and Modelling	Dr. Aniket Bhalkikar	Resource person for SAP2000 sessions covering modelling workflow, materials, section properties, load assignment, analysis and result interpretation.
QGIS: GIS and Geospatial Applications	Dr. Sovan Sankalp	Resource person for QGIS sessions covering GIS fundamentals, spatial data handling, digitisation, DEM/terrain interpretation and civil engineering applications.

Brief Profiles:

Ms. Neha Anil Choudhari

Ms. Neha Anil Choudhari is a structural design professional with more than 12 years of experience in structural analysis, design and detailing of reinforced concrete structures. She has worked on residential, commercial, institutional and infrastructure-related projects, including multi-storeyed buildings, foot over bridges, hospital buildings, industrial sheds and strengthening/retrofitting assignments. She is proposed as the resource person for the AutoCAD module of the training programme.

Area	Details
Core Expertise	Structural analysis and design of reinforced concrete structures; structural detailing; building drawing interpretation; high-rise structural design support; drawing coordination and site coordination.
Professional Experience	Freelancer Structural Design Engineer since 2019; previously Structural Design Engineer at SR Consultant, Aurangabad; Assistant Professor at Vidya Jyothi Institute of Technology, Hyderabad; Structural Engineer at My Home Constructions Pvt. Ltd., Hyderabad; Trainee Structural Engineer at Sumatec Inc., Aurangabad.
Project Exposure	Residential buildings up to G+5 as a freelancer; G+8 residential buildings; commercial centres; hospital buildings; industrial sheds; foot over bridges; strengthening of existing school building; high-rise projects up to 36 floors during earlier industry experience.
Software Skills	AutoCAD, ETABS, STAAD-Pro, SAFE, SAP2000, MS Office and MATLAB.
Education	M.E. in Structural Engineering, Government College of Engineering, Aurangabad; B.E. in Civil Engineering, Marathwada Institute of Technology, Aurangabad.

Dr. Aniket Bhalkikar

Dr. Aniket Bhalkikar is a Civil Engineering academic and structural engineering specialist with experience in building design and analysis, seismic safety assessment, rapid visual survey, risk assessment and structural health monitoring. He is proposed as the SAP2000 resource person for the structural analysis and modelling component of the training programme.

Area	Details
Core Expertise	Structural design and analysis; seismic safety assessment of existing buildings; rapid visual survey; risk assessment; structural health monitoring; applications of AI and ML in structural engineering.
Education	Ph.D. in Civil Engineering, International Institute of Information Technology Hyderabad; M.Tech in Computer Aided Structural Engineering, IIIT Hyderabad; B.E. in Civil Engineering, Marathwada Institute of Technology, Aurangabad.

Area	Details
Academic Experience	Assistant Professor at CHRIST (Deemed to be University), Bengaluru; taught undergraduate and postgraduate civil engineering and architecture courses including Design of Steel Structures, Design of High Rise Structures, Basics of Civil Engineering and Engineering Mechanics, and Logic of Structures.
Consultancy / Industry Engagement	Design and detailing of RC retaining walls; design and detailing of G+2 RC building for railway facilities; instrumentation of railway over-bridge for condition/structural health monitoring.
Software Skills	SAP2000, ETABS, AutoCAD, QGIS, MATLAB, Python, MS Excel and Excel VBA. Developed SAP2000 API-based Excel VBA automation for modelling and analysis workflows.

Dr. Sovan Sankalp

Dr. Sovan Sankalp is a Water Resources Engineer with expertise in hydrology, GIS and AI/ML. He has experience in hydrological analysis, groundwater systems, urban flood modelling and geospatial intelligence, with strong integration of GIS, remote sensing and AI/ML techniques.

Area	Details
Current Position	Postdoctoral Fellow at AHRC, IIT Bhubaneswar.
Core Expertise	Hydrological analysis and modelling; groundwater hydrology and assessment; urban flood analysis and risk mapping; remote sensing and GIS applications; AI/ML-based time series forecasting; drought monitoring; urban climate and impervious surface analysis.
Software Skills	HEC-RAS; HEC-HMS; CCHE-2D; CES; ArcGIS Pro; ENVI; ERDAS IMAGINE; Google Earth Engine; Python; R; MATLAB; SPSS; AutoCAD.
Experience	10+ years of research, teaching and project experience in Water Resources Engineering; former Associate Professor, Civil Engineering, Gandhi Institute for Education and Technology, Odisha; former Assistant Professor, Department of Civil Engineering, Centurion University of Technology and Management, Odisha.
Education	Ph.D. in Civil Engineering (Water Resources Engineering), NIT Rourkela; M.Tech in Civil Engineering (Water Resources Engineering), NIT Rourkela; B.Tech in Civil Engineering, SOA University, Odisha.

3. Rationale of the Programme

Civil engineering practice is increasingly supported by digital tools for drafting, structural modelling, analysis, mapping, planning and project documentation. Students entering internships, academic projects and early career opportunities are expected to have basic exposure to industry-relevant software workflows.

This programme is designed to provide II and III Year Civil Engineering students with guided hands-on exposure to AutoCAD, SAP2000 and QGIS. The focus is not on advanced design, but on basic software workflow, drawing interpretation, model preparation, result interpretation, map preparation and technical documentation.

4. Suitability for II and III Year Students

The programme is appropriate for a combined batch of II and III Year Civil Engineering students. II Year students will benefit from early exposure to drawing, structural systems and mapping, while III Year students will gain stronger readiness for design courses, internships and project work.

Student Level	Training Emphasis
II Year Students	Basic interface, drawing reading, simple drafting, basic modelling, map preparation and interpretation
III Year Students	Deeper interpretation of drawings, load application, analysis outputs, thematic mapping and case-based applications

5. Programme Objectives

1. Provide students with practical exposure to commonly used civil engineering software tools.
2. Develop foundational skills in structural drafting, structural modelling, analysis interpretation and geospatial mapping.
3. Improve students' ability to read, prepare and interpret basic engineering drawings.
4. Introduce the workflow of structural modelling and analysis using SAP2000.
5. Familiarise students with GIS-based mapping and applications in civil engineering.
6. Strengthen industry readiness through guided hands-on exercises, assignments and project report preparation.
7. Support students in preparing for internships, mini-projects, consultancy exposure and entry-level professional work.

6. Hour Distribution

Component	Hours
Direct online/offline training sessions	30 Hours
Reading material, practice and assignments	30 Hours
Total	60 Hours

7. Module-wise Distribution

Module	Topic	Direct Training Hours	Reading/ Assignment Hours	Total Hours	Mode
Module I	Structural Drafting and Detailing using AutoCAD	10	10	20	Online
Module II	Structural Analysis and Modelling using SAP2000	10	10	20	Online
Module III	GIS and Geospatial Applications using QGIS	8	8	16	Offline Practical
Module IV	Project Report Guidance and Final Submission	2	2	4	Online / Offline
Total		30	30	60	

8. Day-wise Training Schedule

The direct training component will be delivered over 30 days. Each day will also include reading material, guided practice or assignment work so that the total engagement becomes 60 hours.

Day	Module	Direct Training Topic	Mode	Reading/ Assignment Component
1	AutoCAD	Introduction to structural drawings and role of AutoCAD in civil engineering	Online	Reading: Types of civil and structural drawings
2	AutoCAD	AutoCAD interface, units, drawing setup and basic commands	Online	Practice: Basic line, rectangle, offset, trim and dimension commands
3	AutoCAD	Layers, line types, text, dimensions and drawing standards	Online	Assignment: Prepare a simple layered drawing template
4	AutoCAD	Grid systems and building layout concepts	Online	Practice: Draw a basic grid layout
5	AutoCAD	Column layout drawing	Online	Assignment: Prepare a simple column layout
6	AutoCAD	Foundation and footing layout concepts	Online	Practice: Draw isolated footing layout
7	AutoCAD	Beam layout and framing plan	Online	Assignment: Prepare a simple beam layout
8	AutoCAD	Slab layout and structural sections	Online	Practice: Prepare one simple structural section
9	AutoCAD	Introduction to reinforcement detailing and BBS	Online	Assignment: Prepare a simple reinforcement detail / BBS format
10	AutoCAD	Review exercise and discussion	Online	Submission: One simple structural drawing sheet
11	SAP2000	Introduction to structural analysis software and SAP2000 workflow	Online	Reading: Basic structural modelling terminology
12	SAP2000	SAP2000 interface, units and model creation	Online	Practice: Create a simple model file
13	SAP2000	Definition of materials and section properties	Online	Assignment: Define basic material and member properties
14	SAP2000	Supports, joints, members and structural idealisation	Online	Practice: Model a simple beam or frame
15	SAP2000	Load types in buildings	Online	Reading: Dead load, live load and basic load concepts
16	SAP2000	Load assignment and basic load combinations	Online	Assignment: Apply loads to a simple model
17	SAP2000	Running analysis and checking model behaviour	Online	Practice: Run analysis and observe model response



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Day	Module	Direct Training Topic	Mode	Reading/ Assignment Component
18	SAP2000	Interpretation of displacement, shear force, bending moment and axial force	Online	Assignment: Record and interpret basic output values
19	SAP2000	Introduction to design checks and safe interpretation of software results	Online	Reading: Importance of engineering judgement in software use
20	SAP2000	Simple case study and review discussion	Online	Submission: Screenshot of model and short interpretation
21	QGIS	Introduction to GIS and civil engineering applications	Offline Practical	Reading: Applications of GIS in civil engineering
22	QGIS	QGIS interface, project setup and data loading	Offline Practical	Practice: Load base map and create project file
23	QGIS	Coordinate systems, projections and georeferencing basics	Offline Practical	Assignment: Identify CRS and map coordinates
24	QGIS	Digitisation of engineering features	Offline Practical	Practice: Digitise roads, buildings, drains or water bodies
25	QGIS	Attribute table, symbology and map styling	Offline Practical	Assignment: Prepare a simple thematic map
26	QGIS	DEM, contour and terrain interpretation	Offline Practical	Practice: Generate or interpret terrain / contour map
27	QGIS	Civil engineering applications of GIS	Offline Practical	Assignment: Prepare site / infrastructure / watershed map
28	QGIS	Map layout, legend, scale, north arrow and PDF export	Offline Practical	Submission: One final QGIS map in PDF format
29	Project Report	Project report structure, topic finalisation and group mentoring	Online / Offline	Draft: Prepare report outline and collect outputs
30	Project Report	Final report submission review and programme closure	Online / Offline	Submission: Group project report in PDF format



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9. Module Details and Expected Learning Outcomes

Module I: Structural Drafting and Detailing using AutoCAD

Duration: 10 direct hours + 10 hours reading/practice/assignment

Learning focus: Basic structural drawing conventions, drawing setup, layers, dimensions, grid layout, column layout, footing layout, beam layout, slab/section details and introductory reinforcement detailing.

- Use basic AutoCAD commands for civil engineering drafting.
- Prepare simple structural layouts such as grid, column, beam and footing layout.
- Understand basic drawing standards, annotation and reinforcement detailing conventions.
- Submit one simple structural drawing sheet in PDF format.

Module II: Structural Analysis and Modelling using SAP2000

Duration: 10 direct hours + 10 hours reading/practice/assignment

Learning focus: Basic structural modelling workflow, material and section definition, supports, joints, members, load application, load combinations, analysis run and interpretation of displacement and member force outputs.

- Create a simple beam, truss or frame model in SAP2000.
- Assign basic material properties, section properties, supports and loads.
- Run a basic structural analysis and interpret output at an introductory level.
- Submit model screenshots and a short result interpretation.

Module III: GIS and Geospatial Applications using QGIS

Duration: 8 direct hours + 8 hours reading/practice/assignment

Learning focus: GIS concepts, QGIS interface, coordinate systems, spatial data handling, digitisation, attribute table, symbology, DEM/contour interpretation, infrastructure mapping and map layout export.

- Use QGIS to create and manage simple spatial datasets.
- Digitise roads, buildings, drains, water bodies or other civil engineering features.
- Prepare basic thematic, infrastructure or terrain maps.
- Submit one QGIS map in PDF format.

Module IV: Project Report Guidance and Final Submission

Duration: 2 direct hours + 2 hours report preparation

Learning focus: Topic selection, report structure, documentation of software workflow, inclusion of screenshots/drawings/maps/results, interpretation and final report submission.

10. Project Report Component

As part of the 60-hour training programme, students will submit a group-based project report at the end of the training. The project report will be based on the hands-on sessions conducted using AutoCAD, SAP2000 and QGIS.

Students may choose any one topic from the suggested list, depending on their interest and the software platform they are comfortable with. The project is intended to consolidate the learning from the training programme and help students demonstrate basic industry-oriented software skills.

Particulars	Details
Type of Submission	Group-based project report
Group Size	4-5 students per group
Expected Number of Groups	Approximately 14-17 groups for 70 students
Software Choice	AutoCAD / SAP2000 / QGIS
Report Length	8-12 pages, excluding annexures
Submission Format	PDF report with necessary drawings, screenshots, maps, model views or analysis outputs
Presentation	Not required

11. Suggested Project Topics

A. AutoCAD-Based Project Topics

1. Preparation of Structural Layout for a Small Residential Building - grid layout, column layout, beam layout and basic structural drawing sheet.
2. Column and Footing Layout for a Simple Building - column positions, footing layout and basic foundation drawing.
3. Beam and Slab Layout of a Classroom / Small Hall - beam layout, slab panel identification and structural plan.
4. Reinforcement Detailing of a Simple RCC Beam - beam reinforcement drawing with basic bar details.
5. Preparation of a Simple Bar Bending Schedule - BBS for a simple beam, slab or footing based on a given drawing.

B. SAP2000-Based Project Topics

1. Analysis of a Simply Supported Beam using SAP2000 - modelling, load application, shear force, bending moment and deflection interpretation.
2. Analysis of a 2D Portal Frame - support assignment, loading and interpretation of member forces.
3. Comparison of Beam Behaviour under Different Loading Conditions - point load, uniformly distributed load and combined loading.
4. Analysis of a Simple Truss Structure - modelling and interpretation of axial forces in members.
5. Basic Structural Modelling of a Small Building Frame - modelling and interpretation of displacement and member force results.

C. QGIS-Based Project Topics

1. GIS-Based Infrastructure Mapping of a Campus / Local Area - roads, buildings, open spaces, drainage lines and infrastructure features.
2. Road Network and Accessibility Mapping - road network map, access points, junctions and connectivity.
3. Watershed and Drainage Mapping using QGIS - drainage network, contour map and basic terrain interpretation.
4. Land Use / Land Cover Mapping of a Selected Area - built-up area, vegetation, water bodies, open land and roads.
5. Site Suitability Mapping for a Civil Engineering Project - road access, slope, land use, water body proximity and available land.

13. Suggested Project Report Format

1. **Title Page:** Project title, student names, registration numbers, department, institution and software used.
2. **Introduction:** Brief background of the selected topic and its relevance to civil engineering.
3. **Objectives of the Project:** Two to four clear objectives.
4. **Software Used:** Brief note on AutoCAD / SAP2000 / QGIS, depending on the selected project.
5. **Methodology:** Step-by-step explanation of how the work was carried out.
6. **Inputs/ Data Used:** Drawing data, load assumptions, site details, map data, dimensions or other input used.
7. **Work Done Using the Software:** Description of the drawing, model, map or analysis prepared.
8. **Software Outputs:** Screenshots, drawings, model views, maps, analysis results or exported PDF sheets.
9. **Results and Interpretation:** Simple explanation of the output generated from the software.
10. **Learning Outcome:** What the students learned and how it is useful for civil engineering practice.
11. **Conclusion:** Brief summary of the project work.
12. **References/ Acknowledgement:** Data source, tutorial, class material or guidance used.
13. **Annexure:** Additional screenshots, drawings, maps, tables or calculations, if required.

14. Expected Project Deliverables

1. Project report in PDF format.
2. Software outputs included within the report, such as AutoCAD drawing, SAP2000 model screenshots, analysis results or QGIS maps.
3. One individual contribution note from each student, limited to 100-150 words, included at the end of the report.

15. Evaluation Pattern

Component	Marks
Attendance and participation	20
Module-wise assignments and practice submissions	30
Final group project report	40
Individual contribution note and documentation discipline	10
Total	100

16. Project Report Evaluation Rubric

Component	Marks
Relevance and clarity of selected topic	10
Correct use of software workflow	25
Quality of output: drawing / model / map	25
Interpretation of results	15
Report structure and documentation	15
Teamwork and individual contribution note	10
Total	100

17. Certification Criteria

Certificates may be awarded to students who satisfy the following requirements:

1. Minimum 75% attendance.
2. Completion of module-level assignments as prescribed by the trainers.
3. Submission of the final group project report in PDF format.

18. Expected Outcome

At the end of the 60-hour training programme, students will gain introductory hands-on exposure to AutoCAD, SAP2000 and QGIS. The programme will help them understand basic civil engineering digital workflows related to drafting, structural modelling, analysis interpretation and geospatial mapping. It will also support their preparedness for internships, academic projects, field documentation and industry-oriented learning.

19. Conclusion

The proposed training programme offers a balanced and manageable exposure to three important software platforms used in civil engineering practice. The design of the programme is suitable for II and III Year Civil Engineering students because it combines guided software demonstration, simple hands-on practice, reading material, assignments and a final project report. The final report-based submission will allow students to document their learning without the additional burden of a formal presentation.